

Computer Department			
Program: Computer Science	Final Practical Take Home Exam 2 nd Semester	Date : 12/06/2020	
Major : CSE-2		Time : -	
Course : Programming languages		Instructor: Dr. Suheir Harb	

Suppose the weekly hours for all employees are stored in a two-dimensional array. Each row records an employee's seven-day work hours with seven columns. For example, the following array stores the work hours for eight employees.

	Su	M	T	W	Th	F	Sa	Su
Employee1	0	2	1	3	4	5	2	2
Employee2	1	7	3	4	3	3	4	4
Employee3	2	3	3	4	3	3	2	2
Employee4	3	9	3	4	7	3	4	1
Employee5	7	8	6	8	6	8	7	8
Employee6	5	3	4	4	6	3	4	4
Employee7	6	3	7	4	8	3	8	4
Employee8	7	6	3	5	9	2	7	9

Write a program that

- Prompt the user to enter number of employees
- Prompt the user to enter the employees names and save their names in a one dimensional array. (Check for availability if name exist, no duplicate names)
- For each employee, enter the seven-day work hours and save them in a two dimensional array.
- Compute the total hours worked by each employee and save them in one dimensional array.
- Compute the wages of employees as follows:
 - up to 40 hours per week the hour price is 20
 - otherwise the overtime hour price is 1.2 regular hour price.
- Design a menu to the user so that he can choose from as follows where the menu is presented to user as long as he does not select 5:
 1. Display employee
 - User entered employee name, Search for employee by name and display his wage and total hours
 2. Display employees
 - Show employee names in decreasing order of the total hours and their wages
 3. Display most active employee
 - Employee with highest total hours
 4. Display average wage and number of employees above average.
 5. Display work statistics
 - 3 categories of total hours {0-19, 20-39, 40-49, >=50}
 - Count the number of employees in each category & Display total hours distribution as bar chart, total Hour distribution for the sample data given above is:
 - 0-19 *
 - 20-39 ****
 - 40-49 **
 - >=50 *
 6. Quit

Notes:

- Use Stepwise refinement. (break the problem into smaller manageable sub problems)
- Use methods in your code (modular code), some of them are listed here for your benefit:

int[] calTotalHrs(int[][] hrs)	double[] calWages(int[] tHrs)	printStr(String[] str)
double avgWage(double[] wages)	int aboveAvg(double[] wages, double avg)	boolean search(String name)
int [] hoursDist(int[] tHrs)	String[] barChart(int[] hDist)	int activeEmployee(int[] tHrs)

End of question